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Hospital Data warehouse :

a) Three classes of schemas

* Star schema

* Snowflake schema

* fact constellation (Galaxy) schema

b) Star schema

fact table : fee-fact

* Day-ID

* Month-ID

* Year-ID

* Doctor-ID

* Hospital-ID

* Patient-ID

* Count

* Charge

Dimension table :

* Time (Day, Month, Year)

* Doctor

* Hospital

* Patient

SQL Query :

Select Day, Month, Year, Doctor, Hospital

Patient, count charge

FROM Fee;

2) Big university Data warehouse :

a) star schema

fact table : Grade - fact

* Student - ID

* Course - ID

* Semester - ID

* Instructor - ID

* Count

* Avg - grade

Dimension tables :

* Student

* Course

* Semester

* Instructor

6) OLAP operations :

Roll up : Semester $\rightarrow$ Year

Slice : Course = CS

Dice : Select required semester / year if needed

c) No. of Cuboids :

student = 5

Course = 5

Semester = 5

Instructor = 5

Total Cuboids = $5 \times 5 \times 5 \times 5 = 625$

Weather Data Warehouse :

star schema

fact table : weather-fact

* Station-ID

* Location-ID

* Time-ID

* Air - Pressure

* Temperature

* Precipitation

Dimension table :

* weather station

* location

* time.

4) Data cube :

4) Given

* n dimensions

* each dimension has p values

a) Maximum cells in base cuboid

$$p^n$$

b) Minimum cells in base cuboid

$$p^n$$

c) Maximum cells in data cube

$$(p+1)^n$$

d) Minimum cells in data cube

$$p^{n+1}$$

5) University academic Performance Data Warehouse :

a) Star schema

* Student-ID (FK)

* Course-ID (FK)

* Faculty-ID (FK)

* Time-ID (FK)

* Total marks

* Attendance - Percentage

Dimension tables :

Student : Department, Year of study, Gender.

Course : program, Semester, subject

Faculty : Department, Designation

Time : Year, Semester, Month.

b) OLAP operations :

To display department-wise average marks for each academic year :

Roll up : Student $\rightarrow$ Department

Roll up : Time $\rightarrow$ Year

Aggregate : calculate Average (Total-marks)

c) Total No. of cuboids :

formula : $4 \times 3 \times 3 \times 5 = 180$ Cuboids

Total = Level student $\times$ Level in course $\times$ Level
in faculty $\times$ Levels in time